CBOT SOYBEAN MEAL FUTURES (ZM)

Spread & Curve Structure — Fundamental Reference

The Crush, Global Livestock Demand, and Protein-Meal Substitution

This document presents a fact-based reference on the fundamental drivers of CBOT Soybean Meal futures (ticker: ZM), contract size 100 short tons, the second of the two co-products (alongside soybean oil, ZL) generated by crushing soybeans, and — by weight and by historical share of total crush revenue — the larger of the two. Where soybean oil's demand profile has been transformed in recent years by the rapid growth of biofuel feedstock demand (described in the Soybean Oil reference document in this series), soybean meal's demand fundamentals remain anchored overwhelmingly in its traditional role as the dominant protein source in global livestock and poultry feed rations. This makes soybean meal the contract in the soybean complex most directly and continuously linked to the livestock sector dynamics described in the Live Cattle and Lean Hog reference documents elsewhere in this series, and to the global protein-meal complex in which soybean meal competes with, and is priced relative to, other major protein meals including canola meal, cottonseed meal, and fishmeal. This document presents the soybean meal-specific fundamentals in full, with particular emphasis on the crush relationship to soybeans and oil, the global livestock feed demand structure, Argentina's outsized role as the world's leading meal exporter, and the protein-meal substitution relationships that distinguish soybean meal's demand framework from soybean oil's.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and the Crush Relationship

CBOT Soybean Meal is one of the two co-products of the soybean crush (alongside Soybean Oil, ZL), priced and traded as its own dedicated futures contract, and — reflecting its larger output weight per bushel crushed — the higher-volume of the two co-product contracts in terms of underlying physical tonnage represented.

Parameter	Detail
Exchange	CME Group — Chicago Board of Trade (CBOT)
Ticker	ZM (front month)
Underlying	Soybean meal, 48% protein (the standard "high-pro" specification; also referred to in the cash market as "soybean meal, 48% protein, minimum 0.5% fat")
Contract size	100 short tons (approximately 90.7 metric tonnes)
Quotation	US dollars per short ton ($/short ton); minimum tick = $0.10/short ton = $10.00 per contract
Contract months	January (F), March (H), May (K), July (N), August (Q), September (U), October (V), December (Z) — eight delivery months per year, the same dense calendar structure as Soybean Oil (ZL), denser than the underlying Soybean (ZS) contract's seven months.
Settlement type	Physical delivery via shipping certificate at exchange-approved soybean meal storage and loading facilities, predominantly in the Illinois/Chicago-area processing network — a bulk dry-product handling system distinct from both the grain-elevator system (ZS, ZC, ZW) and the liquid-tankage system (ZL) described elsewhere in this series.
Delivery grade	48% protein soybean meal meeting exchange-specified quality parameters (protein content, fibre, moisture); a lower-protein "44% meal" or "regular meal" specification also exists in the cash/physical market (Section 1.1) but is not the ZM futures delivery standard.
Trading hours	CME Globex: Sunday–Friday 19:00–07:45 ET and 08:30–13:20 ET, the same general electronic-session structure as the other soybean-complex contracts.
Last trading day	Business day prior to the 15th calendar day of the delivery month
Contract value at $350/short ton	$35,000 per contract
Related contracts	Soybeans (ZS) and Soybean Oil (ZL) — together with ZM, these constitute "the soybean complex" and the three legs of the board crush described in Section 3 and in the CBOT Soybeans and Soybean Oil reference documents.

1.1 48% Protein vs. 44% (Regular) Meal — Why the Futures Standard Matters

- Soybean meal is produced and sold commercially in more than one protein specification, with 48% protein ("high-pro," the ZM futures delivery standard) and 44% protein ("regular meal," a lower-protein specification that includes a higher proportion of soybean hull material) both actively traded in the physical market. The protein-content difference reflects how much hull material is removed during processing — higher-protein meal has had more hull removed, yielding a more concentrated protein product but a correspondingly smaller total meal output per bushel crushed. Because ZM prices the 48% standard specifically, the cash-market spread between 44% and 48% meal is a basis-level consideration for participants whose physical exposure is to the lower-protein specification, requiring a basis adjustment when hedging via ZM futures.

2. Why Soybean Meal Is a Co-Product, Not a Standalone Crop

As with soybean oil, soybean meal has no independent agricultural production process of its own — it is generated entirely as one of two outputs of crushing soybeans, meaning its supply-side fundamentals are inherited directly from the soybean crop and crush-capacity fundamentals described in full in the CBOT Soybeans reference document.

2.1 The Fixed-Ratio Yield — Meal as the Larger Co-Product

- Crushing a 60 lb bushel of soybeans yields approximately 44–48 lb of soybean meal and 11 lb of crude soybean oil (plus hulls and processing losses), meaning meal represents the larger share of total physical output by weight from every bushel crushed — the inverse framing of the same fixed-ratio relationship described from the oil side in the Soybean Oil reference document. As with oil, this ratio is fixed by the biology of the soybean and cannot be substantially altered on any near-term production-planning timescale, meaning meal supply growth is, like oil supply growth, fundamentally a function of total soybean crush volume rather than of any meal-specific production decision.

2.2 Historical Revenue Share — Meal as the Traditional Anchor Product

- As referenced in the Soybean Oil reference document's discussion of the crush margin, soybean meal has historically represented the larger share of total crush revenue, reflecting both its larger output weight per bushel and the scale of global livestock and poultry feed demand for protein meal. While the rapid growth of biofuel-driven oil demand has, at various points in recent years, narrowed the gap in relative revenue contribution between the two co-products, meal's role as feed-protein demand's primary agricultural source remains the more traditional, longer-established demand anchor for the overall soybean processing industry, relative to oil's comparatively newer and more rapidly evolving biofuel-driven demand profile.

3. The Crush — Revisited From the Meal Side

The crush relationship between soybeans, meal, and oil is described in detail in the CBOT Soybeans reference document, with the oil-specific perspective covered in the Soybean Oil reference document. This section summarises the relationship from the meal leg's specific perspective.

3.1 The Board Crush

- As described in both companion reference documents, the crush margin — combined meal and oil revenue per bushel crushed, less the soybean input cost — can be calculated directly from the three CBOT contract prices (ZS, ZM, ZL) using the fixed yield ratios described in Section 2.1. Processors hedge their actual physical crush margin using offsetting ZS/ZM/ZL futures positions, and the board crush is also directly traded by speculative participants taking a view on processing economics independent of any single leg's flat-price direction. Given meal's larger weight contribution to the crush calculation, meal price movements typically exert a somewhat larger mechanical influence on the calculated board crush margin than an equivalent percentage move in oil, all else equal — a direct consequence of the output-weight asymmetry described in Section 2.1.

3.2 NOPA Monthly Crush Report — Meal Production and Stocks

- As described in the Soybean Oil reference document, the National Oilseed Processors Association (NOPA) publishes a monthly crush report covering soybean processing volume, and meal and oil production and stocks, roughly two weeks after each month-end. The meal-specific production and stocks figures in this report are tracked by livestock feed market participants as a higher-frequency, industry-sourced complement to USDA's own monthly and WASDE-embedded meal supply estimates.

4. Global Livestock Feed Demand — The Primary Demand Pillar

Unlike soybean oil's increasingly bifurcated demand structure (food use plus a rapidly growing biofuel pillar, described in the Soybean Oil reference document), soybean meal's demand remains concentrated overwhelmingly in a single application: as the dominant protein source in livestock and poultry feed rations, both domestically in the United States and globally.

4.1 Why Soybean Meal Dominates as a Feed Protein Source

- Soybean meal's amino acid profile — particularly its relatively high and well-balanced essential amino acid content, including lysine, a frequently limiting amino acid in cereal-grain-based feed rations — makes it an unusually effective and cost-competitive protein source for monogastric livestock (pigs and poultry specifically, whose single-chambered digestive systems require a more concentrated and complete amino acid profile than ruminants like cattle), in combination with its large-scale, cost-competitive global production base described in the CBOT Soybeans reference document's production geography section.
- **Complementarity with corn in feed rations**: as described in the Corn reference document's feed-demand section, corn provides the energy (caloric) component and soybean meal provides the protein component of most US (and much of global) livestock and poultry feed formulation, making the two ingredients economic complements rather than substitutes in most feed rations — a different relationship from the corn-wheat feed substitution relationship described in the Wheat reference document, where corn and feed-wheat are substitutable energy sources rather than complementary energy-and-protein components.

4.2 The Poultry and Hog Sector Connection

- Because monogastric livestock (hogs and poultry) rely more heavily on a concentrated protein meal source than ruminants (cattle, which can derive a meaningful share of protein from forage and roughage in addition to concentrate feed), soybean meal demand is structurally more closely tied to the hog and poultry sectors specifically than to the cattle sector, in contrast to corn, whose feed demand is more evenly distributed across all three major livestock categories.
- **The hog cycle**: as described in the Lean Hog reference document, US (and global) hog production responds relatively quickly to feed-cost and hog-price profitability signals given the shorter hog production cycle relative to cattle, making hog sector soybean meal demand somewhat more responsive to near-term meal price changes than the cattle sector's comparatively modest and slower-moving meal demand.
- **Poultry production cycles**: poultry (broiler and layer) production cycles are shorter still than hog production cycles, making poultry-sector feed formulation decisions — and therefore poultry-sector meal demand — the most responsive of the three major livestock categories to near-term relative feed-ingredient pricing.

4.3 USDA Inventory Reports as Indirect Meal-Demand Indicators

- As referenced in the Corn reference document, USDA's quarterly Hogs and Pigs report and semi-annual Cattle inventory report (plus, relevant specifically here, USDA's broiler and egg-sector production statistics) provide the herd/flock-size data that translates into the feed-protein-demand component of the soybean meal balance sheet, making these livestock-sector reports indirectly but significantly relevant to ZM even though they are not meal-specific publications.

5. Global Production and Trade — Argentina's Distinctive Role

Soybean meal supply geography mirrors the underlying soybean production geography described in the CBOT Soybeans reference document (United States, Brazil, Argentina), but with one structurally important divergence: Argentina's export profile is weighted overwhelmingly toward processed meal and oil rather than raw soybeans, making Argentina the world's leading exporter of soybean meal specifically, a different ranking from its position in raw soybean production and export.

Country	Role in Global Meal Trade	Mechanism	ZM Relevance
Argentina	The world's leading soybean meal exporter by a wide margin	Argentina's long-standing export tax policy applies higher rates to raw soybeans than to processed meal and oil, an explicit policy choice favouring domestic value-added processing (detailed in the CBOT Soybeans reference document); this has built a very large Argentine domestic crushing industry oriented toward export	Argentine crush capacity utilisation, weather affecting the Argentine soybean crop, and Argentine export tax/policy changes are the single most important non-US variable for the global meal balance specifically — more directly relevant to ZM than to ZS
Brazil	A major meal exporter, though proportionally weighted more toward raw soybean export than Argentina	Brazil's large crush capacity processes a substantial share of its soybean crop domestically, split between meal/oil export and raw soybean export depending on relative economics and demand	Brazilian crush capacity utilisation and the relative attractiveness of crushing versus raw export affect the global meal supply alongside Argentina's more meal-export-concentrated profile
United States	A significant meal exporter, though a smaller share of total US soybean crush output is exported as meal relative to Argentina's export-oriented crush industry	A large share of US-crushed meal is consumed domestically by the substantial US livestock and poultry sector	US domestic meal demand (Section 4) is the dominant call on US crush output; US meal export volume is a smaller, though still relevant, balance-sheet component
European Union	A major meal importer	Limited EU domestic soybean production makes the EU structurally reliant on imported meal, sourced from the US, Brazil, and Argentina	A major demand-side variable; EU import policy (including periodic debates over GM crop import approval, relevant given the predominance of GM soybean varieties in US, Brazilian, and Argentine production) is a tracked policy variable
Southeast Asia and other importing regions	Significant and growing meal importers	Rapidly growing regional poultry and aquaculture sectors (the latter relevant given soybean meal's growing use in aquaculture feed formulation, Section 6.3) driving import demand growth	A structurally growing demand base, though individually smaller than the EU import market

5.1 Why Argentina's Export Tax Structure Created This Divergence

- As described in the CBOT Soybeans reference document, Argentina has historically applied differential export tax rates — higher on raw soybeans, lower on processed meal and oil — specifically to encourage domestic value-added processing rather than raw commodity export. Over time, this policy has resulted in Argentina building substantially more soybean crushing capacity, relative to its own soybean production, than either the US or Brazil, making Argentina disproportionately important to the global meal (and oil) balance specifically, even though its share of total world soybean production is considerably smaller than the US or Brazilian share.

- **The practical consequence for ZM specifically**: an Argentine drought or other production shortfall (such as the well-documented severe drought affecting the 2022/23 Argentine crop, referenced in the CBOT Soybeans reference document) has a proportionally larger effect on the global meal and oil balance than its effect on the global raw soybean balance alone would suggest, precisely because Argentina's crush-oriented export structure means a smaller Argentine crop disproportionately reduces the volume available for the meal- and oil-export channels specifically.

6. The Global Protein Meal Complex — Cross-Meal Substitution

As with soybean oil's position within the broader vegetable oil complex (described in the Soybean Oil reference document), soybean meal is one of several protein meals that compete, to varying degrees, for substitutable livestock feed demand, making the broader protein meal complex a relevant cross-market consideration, albeit one in which soybean meal's dominance is somewhat more pronounced than soybean oil's position within the vegetable oil complex.

Protein Meal	Relative Scale vs. Soybean Meal	Primary Geography/Source	Substitution Relevance
Canola/rapeseed meal	A significant secondary protein meal, though considerably smaller in global volume than soybean meal	Canada, European Union (a co-product of canola/rapeseed oil crushing, mirroring the soybean crush relationship)	A partial substitute in some livestock rations, particularly in regions with significant canola crushing capacity (Canada, EU); lower protein content than soybean meal generally limits full substitutability
Cottonseed meal	A smaller-volume protein meal, a co-product of cottonseed crushing	Produced in cotton-growing regions globally (US, China, India, and others)	A regional/local substitute where cottonseed crushing capacity exists; gossypol content (a naturally occurring compound in cottonseed) limits its use in some monogastric rations without processing to reduce gossypol levels
Sunflower meal	A regionally significant protein meal, particularly in regions with substantial sunflower oil crushing	Ukraine, Russia, EU — broadly the same producing geography as sunflower oil	A partial regional substitute, particularly in European feed formulation; affected by the same Ukraine war-related supply disruption referenced for sunflower oil
Fishmeal	A smaller-volume but higher-value specialty protein meal	Peru and Chile (anchoveta-based production) are the dominant global sources	Used predominantly in aquaculture and some specialty livestock/pet food applications rather than as a broad substitute for soybean meal in mainstream livestock rations; Peruvian anchoveta catch quotas (affected by El Nino-driven population fluctuations) are a tracked supply variable for this specific niche
Distillers dried grains with solubles (DDGS)	A significant volume co-product, though lower in protein concentration than soybean meal	United States (a co-product of corn ethanol production, described in the Corn reference document)	A partial substitute for a portion of both corn (energy) and soybean meal (protein) in feed rations, given DDGS's combined energy-and-protein content; DDGS availability is directly tied to US ethanol production volume, creating an indirect link between corn ethanol policy and soybean meal demand at the margin

6.1 Why Soybean Meal Remains the Dominant Protein Meal Despite Competition

- Soybean meal's combination of relatively high and well-balanced protein content (Section 4.1), large-scale and broadly distributed global production (US, Brazil, Argentina), and well-established global trading and logistics infrastructure gives it a structural cost-competitiveness and reliability advantage that has sustained its dominant market position despite the existence of multiple competing protein meal sources, none of which individually approaches soybean meal's combination of scale, protein quality, and production-geography diversification.

6.2 The Soybean Meal-Canola Meal Spread

- Given canola meal's position as the most significant secondary protein meal in markets where both are available (particularly Canada and parts of the EU), the soybean meal-canola meal price spread, adjusted for the protein-content difference between the two, is tracked by feed formulators in these markets as a relative-value signal analogous in function (though smaller in global market significance) to the corn-wheat feed substitution spread described in the Wheat reference document.

6.3 Aquaculture — A Growing Demand Category

- Global aquaculture (fish and shrimp farming) has grown substantially as a share of global animal protein production over recent decades, and soybean meal has, per industry reporting, captured a growing share of aquaculture feed formulation as an alternative or supplement to fishmeal, reflecting both soybean meal's relative cost-competitiveness and ongoing feed-formulation research aimed at reducing aquaculture's historical dependence on fishmeal, whose supply is constrained by wild-catch fishery limits in a way soybean meal's agricultural production base is not.

7. The Forward Curve Structure

As with soybean oil, ZM's forward curve reflects the same general cost-of-carry framework as other storable agricultural commodities, modulated by the underlying soybean crush cycle (inherited from the CBOT Soybeans calendar) and by the global livestock feed demand cycle described in Section 4.

7.1 Carry Inherited From the Crush Cycle

- As described in the Soybean Oil reference document's parallel discussion, because soybean meal supply is generated continuously by ongoing crush activity rather than by a single discrete annual harvest event, ZM's curve does not exhibit as sharply defined a harvest-driven seasonal pattern as ZS itself; it instead reflects a blend of underlying soybean supply seasonality (via crush input availability and the US-Brazil two-hemisphere calendar) and meal-specific demand-side seasonality tied to livestock feed formulation cycles.

7.2 Livestock Feed Demand Seasonality

- US and global livestock and poultry feed demand for soybean meal is relatively stable across the calendar year compared to the more sharply seasonal weather-driven supply-side patterns described for the underlying crop, since livestock and poultry production (and therefore feed consumption) continues year-round rather than concentrating around a single harvest event — meaning ZM's demand-side seasonality is generally less pronounced than its supply-side (crush-input-driven) seasonality, a useful baseline distinction when assessing which side of the balance is driving a given curve move.

7.3 The Crush Spread's Influence on the ZM Curve Specifically

- As described in the Soybean Oil reference document's parallel section, processor forward crush margin hedging links commercial positioning across ZS, ZM, and ZL simultaneously at corresponding forward dates, meaning ZM curve positioning should be assessed alongside the corresponding points on the ZS and ZL curves rather than in isolation — with the added consideration, given meal's larger weight in the crush calculation (Section 3.1), that ZM moves often exert a larger mechanical pull on the calculated board crush margin than equivalent ZL moves.

8. Macro and Cross-Market Drivers

8.1 The Soybean Crush Cycle (Inherited From ZS)

- As described in Sections 2–3, soybean meal's supply-side fundamentals are inherited from the underlying soybean crop and crush-capacity-utilisation cycle described in full in the CBOT Soybeans reference document — the US planting/growing/harvest calendar, the Brazilian growing season's offsetting six-month supply pulse, Argentine crush-export economics (Section 5.1), and NOPA monthly crush data are all directly relevant background for ZM even though they are not meal-specific variables.

8.2 Livestock Sector Profitability and Herd/Flock Cycles

- As described in Section 4, US and global hog, poultry, and (to a lesser, more indirect degree) cattle sector profitability and herd/flock-size cycles are the primary demand-side driver for soybean meal, connecting ZM to the cattle cycle and hog cycle dynamics described in the Live Cattle and Lean Hog reference documents respectively.

8.3 The Protein Meal Complex

- As detailed in Section 6, canola meal, cottonseed meal, sunflower meal, fishmeal, and DDGS each represent partial substitutes or complements for soybean meal in various regional and species-specific feed formulation contexts, making developments in each of these markets a relevant, if generally secondary, cross-market input to soybean meal demand.

8.4 US Dollar and Export Competitiveness

- As a globally traded, USD-denominated commodity with substantial export demand (particularly from Argentina, given its outsized export-oriented crush industry described in Section 5), soybean meal exhibits the standard dollar-priced-commodity relationship described elsewhere in this document series, with ARS/USD (Argentine Peso) movement specifically relevant to Argentine crush-export economics in the same general mechanism described for BRL, VND, and other producer currencies throughout this series.

8.5 Speculative Positioning — COT Data

- **CFTC COT for CBOT Soybean Meal**: weekly managed money net positioning is tracked, consistent with the positioning framework described throughout this document series, and — as noted in both companion soybean-complex reference documents — is frequently monitored alongside the related ZS and ZL positioning data given the crush-margin relationship connecting all three contracts.

9. Documented Seasonal Patterns

ZM's seasonal pattern, like ZL's, is a composite of the underlying soybean supply seasonality (inherited from ZS, including the US-Brazil two-hemisphere calendar) and the comparatively more stable, less sharply seasonal livestock feed demand cycle described in Section 7.2.

Period	Key Driver	Mechanism	Relevant Contracts
Jan–Feb	US post-harvest crush activity continuing at typically strong utilisation; steady livestock feed demand continuing year-round	Crush margin economics and NOPA monthly meal production/stocks data are the primary near-term signal	January, March contracts
Mar–May	Brazilian and (especially) Argentine harvest confirming the Southern Hemisphere contribution to global meal supply	Argentine crush capacity utilisation and export pace are particularly significant given Argentina's outsized meal-export role	May, July contracts
Mid-year	US new-crop growing-season weather (inherited risk from ZS); continuing steady global livestock feed demand	Agricultural-weather risk dominates the supply side; demand side remains comparatively stable	August, September contracts
Year-end	US harvest-driven crush capacity ramp-up; year-end livestock sector inventory and feed-purchasing patterns ahead of the new marketing year	New-crop crush margin economics; USDA year-end Hogs and Pigs / Cattle inventory data feeding into the following year's feed-demand outlook	October, December, January contracts

As with soybean oil, soybean meal has no agricultural growing season of its own, so its seasonal pattern is less calendar-certain than the underlying grain and oilseed crops covered elsewhere in this series; it is instead a composite signal inherited from the underlying soybean crop calendar (with particular attention to Argentina's harvest and export-policy calendar specifically, given Section 5) plus the comparatively steady, less sharply seasonal global livestock feed demand cycle.

10. Key Data Sources and Release Schedule

Report / Data	Publisher	Frequency	Typical Timing	Primary ZM Curve Impact
WASDE (Soybean Meal component)	USDA WAOB	Monthly	~11th–12th of month, 12:00 ET	All contracts — includes meal-specific domestic use, exports, and ending stocks
NOPA Monthly Crush Report	National Oilseed Processors Association	Monthly	~3rd–5th business day of month	Front contracts — meal production and stocks, a higher-frequency complement to USDA figures
USDA Hogs and Pigs Report	USDA NASS	Quarterly	Per quarterly schedule	C3–C8 — US hog herd size, the most directly meal-demand-relevant livestock inventory report
USDA Cattle Inventory Report	USDA NASS	Semi-annual	Jan and Jul	C6–C12 — a smaller but non-trivial meal-demand input given cattle's more modest protein-meal feed share
USDA Chickens and Eggs / Broiler Hatchery reports	USDA NASS	Monthly	Per monthly schedule	C1–C4 — poultry sector feed demand, the most responsive livestock category to near-term meal pricing
Bolsa de Cereales / Argentine export registration data	Bolsa de Cereales de Buenos Aires; Argentine Ministry of Agriculture	Weekly/monthly	Variable	C2–C8 — Argentine crush-export pace, the single most important non-US meal-specific variable
CONAB (Brazilian soybean component)	CONAB (Brazil)	Monthly	~3rd week of month	C3–C8 — inherited from ZS; Brazilian crush capacity utilisation context
EU import/customs data (soybean meal)	Eurostat / national customs agencies	Monthly	Variable	All contracts — confirms EU import demand pace given the EU's structural meal-import dependence
CFTC Commitments of Traders (Soybean Meal)	CFTC	Weekly	Fri ~15:30 ET (3-day lag)	Positioning / crowding — tracked alongside ZS and ZL given the crush-margin relationship

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CBOT Soybean Meal, like Soybean Oil, has no agricultural production process of its own — its entire supply is generated as a co-product of the soybean crush, meaning its supply-side fundamentals are inherited wholesale from the soybean crop and crush-capacity fundamentals described in the CBOT Soybeans reference document. Unlike soybean oil, however, whose demand profile has been substantially reshaped by the rapid growth of biofuel feedstock demand, soybean meal's demand fundamentals remain anchored overwhelmingly in its traditional role as the dominant global protein source for livestock and poultry feed, connecting it directly to the cattle, hog, and poultry sector dynamics described elsewhere in this document series. Argentina's distinctive export-tax-driven crush industry makes it the world's leading meal exporter and the single most important non-US country-specific variable for the global meal balance specifically.

For spread traders, the most reliable analytical framework treats ZM as a soybean-crush-derived supply exposure (best tracked via the same ZS/Brazilian-and-Argentine-calendar fundamentals described in the CBOT Soybeans

reference document, plus NOPA monthly crush data) combined with a comparatively stable, livestock-cycle-driven demand exposure (best tracked via USDA hog, cattle, and poultry inventory data). The board crush margin remains the primary cross-market relationship connecting ZM to the rest of the soybean complex, while Argentine crush-export economics specifically — more so than for either ZS or ZL — warrant close and continuous attention given Argentina's outsized structural role in global meal trade.